BIG IDEAS MATH • GRADE 7 • CHAPTER 1

Adding & Subtracting Rational Numbers: Signed Stack

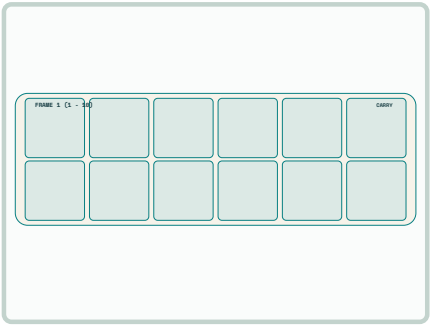
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.7.NS.A.1

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Ten Frame Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Model signed quantities with Two-Tone Zero-Pair Counters (teal +1 face, coral -1 face) and number line jumps.

Sketch Model on Paper:
Draw vector arrows on a number line or circle zero-pairs showing magnitude and sign.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	15	X: 15	Key 15 into X
2.	+/-	X: -15	Negate: X becomes -1
3.	ENTER	Y: -15 | X: -15	Stack lift
4.	8	Y: -15 | X: 8	Key +8 into X

EXPECTED LCD X: -7

1 **Evaluate -28 + 16 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Evaluate -15 - 17 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Evaluate 14.5 - 25.0 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Evaluate -4.5 + (-5.5) on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 1

Adding & Subtracting Rational Numbers: Signed Stack • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Place Value & Two-Tone Zero-Pair Counters (BASE-10)

CAD Blueprint: ten_frame_tray

Target Device: SC-6 / SC-32 Bridge

OpenSCAD Source File:

Education/scad/place_value_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal (+1) & Signal Coral (-1) Two-Tone Counters (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Evaluate -28 + 16 on the stack.** Standard: CCSS.MATH.7.NS.A.1	-12	[2][8] [+/-] [ENTER] [1][6] [+]
2	**Evaluate -15 - 17 on the stack.** Standard: CCSS.MATH.7.NS.A.1	-32	[1][5] [+/-] [ENTER] [1][7] [-]
3	**Evaluate 14.5 - 25.0 on the stack.** Standard: CCSS.MATH.7.NS.A.1	-10.5	[1][4][.][5] [ENTER] [2][5] [-]
4	**Evaluate -4.5 + (-5.5) on the stack.** Standard: CCSS.MATH.7.NS.A.1	-10.0	[4][.][5] [+/-] [ENTER] [5][.][5] [+/-

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for subtracting absolute values without checking which sign had larger magnitude.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.